

CHAPTER FIVE

STORAGE AND MATERIALS HANDLING

Storage Management

Warehouse/store is an essential limb of an industrial unit. It is the depository of all materials required by the industrial unit and supplies materials as and when required. Different types of materials are required for different operations in a production unit. The variety of items stored is so large that a planned system is necessary to keep them safely and in order the stored items should be identified and issued with minimum efforts and minimum time.

Purpose of Stores

Stores play a vital role in the operations of a company. It is in direct touch with the user departments in its day-to-day activities. The most important purpose served by the stores is to provide uninterrupted service to the manufacturing divisions. Further, stores is often equated directly with money, as money is locked up on the stores. The objectives of stores can be classified as follows:

1. To receive raw materials, components, tools, equipments and other items and account for them.
2. To provide adequate and proper storage and preservation to the various items.
3. To meet the demands of the consuming departments by proper issues and account for the consumption.
4. To minimize obsolescence, surplus, and handling.
5. To highlight stock accumulation, discrepancies and abnormal consumption and effect control measures.
6. To ensure good housekeeping so that material handling, material preservation, stocking, receipt and issue can be done adequately.
7. To assist in verification and provide supporting information for effective purchase action.

In general, the main aims of stores management is:

i. Minimization of *Cost of Production*

The main aim of storekeeping function is to minimize the various aspects of costs involved, while providing efficient and effective storekeeping services. Equally important are the costs of materials related to storage itself. These include the cost of capital investment in materials and supplies, the cost of preservation and handling of stores, the cost of record keeping, cost of using of storage equipment and store building, and expenses for arranging light, heat, ventilation, insurance, losses due to spoilage and improper location of stores. Since these costs directly and indirectly influence the production cost, store keeping has to make efforts in reducing them to their minimum level so as to minimize the cost of production.

In this regard, the primary aims of storekeeping are to provide services to the organization by:

- a. Making available a balanced flow of raw materials, equipment, tools and other components to the product line within the least possible time, right quality and quantity.
- b. Providing spare parts, general store, and other materials required by the production department for proper functioning.
- c. Receiving from production unit and issuing to the sales unit the finished products.
- d. Accepting and storing the scrap and other discarded materials and making arrangement for their disposal, in consultation with the appropriate departments and authorities.

ii. *Coordination with other departments*

The store department has to establish coordination with the material control department. But it doesn't mean that it can ignore other departments of the organization. The following departments are closely linked to the stores department.

- | | |
|-------------------------------------|-----------------------------|
| - Inspection Department | - Purchase Department |
| - Design and Engineering Department | - Sales Department |
| - Finance and Accounts Department | - Production Department and |

- Transport Department

The stores department is dependent on each of them for its day-to-day operation. It may not be out of place to mention here that without proper functioning of any one of the departments, the smooth functioning of either the stores department or any other department will come to a standstill; hence the need for a positive correlation between various departments should be the objectives of the stores department.

Storage Systems and Types

Storage Systems

There are two basic systems that can be used in physically controlling stores materials:

- 1) a closed stores system and
- 2) an open stores system

Closed System

A closed stores system is one in which all materials are physically stored in a closed or controlled area. Wherever possible, the general practice is to maintain physical control by locking the storage area. As a rule, no one other than stores personnel is permitted in the stores area. Material enters and leaves the area only with the accompaniment of an authorizing document. This system is designed to afford maximum physical security and to ensure tight accounting control of inventory material.

Open System

The open system is widest use is in highly repetitive, mass production types of operations that exhibit a continuous and predictable demand for the same materials. Most JIT manufacturing systems exemplify this situation.

In plants using the open system, no store room as such exists; each material is stored as close to its point of use as is physically possible. Materials are stored in bins, on shelves, in racks, on pallets, in tote boxes, and so on, much as they would be stored in a storeroom. However, the storage configuration at each work station is arranged to fit the available space. Storage facilities are completely open, and any worker has access to any storage facility.

Random-Access Storage System

The random-access storage system is a unique type of closed stores system, used by a relatively small number of large firms. In this system, no material has a fixed storage location. When an item enters the store room, it is stored in the first available bin or shelf suitable for its storage requirements. When the item is withdrawn from stores, the storage space is available for any other incoming item having similar physical storage requirements. All materials are thus stored at random locations throughout the warehouse.

Types of Stores

In a manufacturing establishment the materials can be classified as raw materials, work-in-progress, bought-out components, spares, consumables and packing materials. In an engineering industry, work-in-progress and bought-out components play an important role, where as in a process industry spare parts will fall into the critical category. Even in a non industrial situation such as in departmental stores and provision stores, materials will be grouped according to their uses such as clothing, utensils, vegetables, fruits, provisions, etc. Depending on the number of items in a major class, it can be split into sub-classes in order to facilitate better control.

Raw Material Stores

The proper control of raw materials is one of the prime factors that influence profitability of a company. These, being the major input into the organization, form the bulk that gets converted into output. A stock out in this class of materials will be immediately reflected by an idle manufacturing line. The stores should be located as near the consumption centers as possible in order to reduce handling.

Work-in-Progress

This class of materials is largely found in engineering industry. In a process industry, these are in the process line itself and hence external storage is not required. The information system required for the control of these should be comprehensive so that the persons in charge know what is where.

Component Stores

In engineering industries, the finished products require complements of the final assembly. These are either manufactured in feeder shops or procured from sub-contractors. In either of these cases they are stored in a component store from where the assembly lines are fed. The stock out of a single component can make the entire assembly line idle.

Spares/Maintenance Stores

The consumption pattern of spares and maintenance items is usually unpredictable. In addition to this one has to contend with the external conflict between the maintenance and stores functions regarding the availability of spare parts and the responsibility for holding these items. In service industries like shipping and transportation and in process industries like chemicals and fertilizers, timely supplies of spares becomes very critical for the stock out cost or the idling cost due to lack of supply is really high. A regular collection, collation and analysis of the consumption data are essential for the proper management of these materials. A common failing is the inadequacy of the existing nomenclature that leads to multi-stocking under different names. This will have to be avoided by appropriate codification schemes.

Tool cribs and Related Stores

Tool cribs are specialties of engineering industries, where the necessary tools fixtures are stored. These can be cutting tools, farm tools, jigs, fixtures, die sets, hand-tools etc. The requirements will be determined by the facilities plan. The management of this store is availability oriented because if a tool is not available a costly machinery will lay idle. Therefore, the tools must be kept in good condition and stored properly.

Consumables

Consumables like cotton waste, grease, lubricating oils etc enable the smooth running of the wheels of the industry. Generally these items do not cause any supply problems but occasionally some can become scarce; petroleum products can be cited as examples especially in stores located in remote areas. Selective control techniques can be applied effectively for these items. Packing materials are also consumables and a general characteristic of these is that they are voluminous. They have to be protected against moisture and excess heat for if they lose their appearance, they cannot be used for marketing purposes.

Finished Goods

Control of finished goods is normally with the marketing or sales department. The peculiarities of the finished goods depend on the type of industry. In a consumer product or a process industry the input to the stock is as per the production schedule while the output will depend on the market conditions. A slack market will lead to stock-piling and a buoyant market will lead to scarcities. The storage space should be adequate to meet the extreme conditions and should protect the product from any deteriorating climatic influences. The shelf life of the product is an important factor that will have to be considered in these cases. In an engineering industry the problems faced are different as the production is usually on the basis of a firm order. Here the sub-assemblies will have to wait till the completion of all other sub-assemblies for erection/assembly at site. Hence the system of control should be oriented towards co-ordination and completion of activities according to a work schedule.

Storage Functions

The following are the activities that storage is expected to perform.

1. Receive all incoming materials and check against quantities and qualities against purchase order, invoices and specifications.
2. Provide adequate and proper storage and preservation.
3. Issue materials against authorized requisitions to production and other departments.

4. See that materials are properly stored against deterioration, theft etc and that they are ready for issue.
5. Minimize obsolescence, surplus, waste and scrap through proper codification, standardization and preservation.
6. Carry out stock verification in accordance with the procedure laid down by the management.
7. Ensure good housekeeping so that material handling, material preservation, stock receipt and stock issue can be done accurately.

These activities and responsibility areas can be put into five functions: receiving, storing, issuing, stock controlling and store documentation. Each of these functions is discussed as follows.

Receiving Materials

The receiving section (function) is the central place where all incoming supplies are received, checked and inspected before storage or use. It is also called Goods Inward Section. Although materials may be received in store room from many sources (internal and external), our emphasis here will be on receiving materials from external sources i.e. from suppliers.

Receiving Procedure

A typical receiving procedure consists of four steps (Dobler and Burt, 2001):

1. *Unloading and checking the shipment.* The number of containers unloaded from the carrier's vehicle is checked against the carrier's manifest (freight bill) to make certain the full consignment has been delivered. All containers are also inspected for external damage; any damage found is inspected by the carrier's representative and noted on the receipt which the receiving clerk signs. Failure to follow this procedure before accepting a shipment can relieve the carrier of all liability, except liability for concealed damage not evident until the container is unpacked.
2. *Unpacking and inspecting the material.* A receiving clerk is held responsible for three verifications. First, he or she checks the material received against the seller's packing slip and against a copy of the firm's purchases order to verify that the correct items have been shipped. Second, the quality of the shipment is verified in the same manner. Finally, the clerk inspects the general condition of the material to determine whether any external damage was incurred during shipment.
3. *Completion of the receiving report.* The paperwork system used varies significantly from firm to firm. In some companies a multipart receiving report form is produced as a by-product when the purchase order is generated. When the receiving clerk has finished the inspection, he or she completes this form by recording the quantities of the items received, indicating those that are still open, and noting any other useful information on the form. In other firms that utilize a computerized purchasing/inventory system, the clerk records the receiving data on the receiving report or a short form, from which a data entry clerk keys the data directly into the computer system. The system then updates the order and inventory records and may also generate a receiving report in either hard copy or electronic form.

Regardless of the system used, four operating groups generally require notification that the material has been received: the requisitioned, the purchasing department, the accounting department, and the inspection department if technical inspection is required.

1. *Delivery of the material.* For non-stock materials, the receiving department is usually responsible for delivery-directly or via an internal delivery service. In the case of inventory materials, the practice varies. In some firms the receiving department is responsible for internal deliveries, while in others this function is performed by an internal transportation service. In still others, stores clerks are responsible for picking up their own materials.

Storing Materials

In this section of the chapter, the various types of storage methods are outlined. The type of equipment to be used in storage depends on the nature of materials to be used, the size and quantity.

1. *Bin*- Bin is a storage equipment with partitions for keeping small items. Each division in a bin is called a pigeon-hole. Generally, only one item is stored in a pigeonhole.
2. *Rack*- Rack is an open type of storage equipment for keeping bulky materials. The number of shelves will depend on the type of materials to be stored, quantity involved etc.
3. *Pallets* - are specially designed platforms for the stacking of materials with a view to the whole load being moved. They are made of either wood or metal depending upon the load they are expected to carry.
4. *Block stacking* - is another version of storage method in which boxes or pallets are stacked directly upon each other.

Stock Issuance

Issue of materials, which is supply of materials from stores to the various users (department) of an organization, is one of the routine activities of the warehouse. Yet, it is very important and has a bearing on the productivity of the departments of the organization, especially of the production and maintenance departments. The user departments often demand immediate issue of the materials for attending to urgent repairs of plant and machinery. Sometimes issues are requested without proper authorization. The warehouse staffs find it difficult to meet such demands due to limited manpower resources.

Stock issuance is executed when the user department comes or sends to the store section and presents a properly authorized issue note-usually referred as stores requisition or issue note or requisition voucher. Store requisition should include the following information:

- Requisitioner
- Description of the material
- Material code
- Quantity required
- Price, and
- Signature of the person authorized to make the request.

Types of Issuance of Materials

i.Capital issues

These are the major capital item (machines, equipments, tools) purchased and issued by the company. In many cases, large capital items will not be held in stock so that purchasing will buy the item and issue it to the users as soon as the item is received at the factory.

ii.Imprest issues

These are items that are required by the user to be constantly on hand. Therefore, the user will need constantly to top-up (to fill) his/her supply.

Iii.Replacement issues

This is when a material has been broken or worn out or obsolete. In this case the authorization for a new issue is the exchange for the old piece of equipment or item.

Iv.Loan Issues

These are usually tools that are required by certain departments for special jobs. The material is to be returned back to the store.

v. Allocated Issues

This is when stock has been allocated by the production plan to a certain operation in a given section. In these cases, the formal production planning document acts as the means of authorization.

vi. General Issues

These are general items of stock needed by various departments within the operation for the day-to-day running of the company.

vii. Special Issues

In cases where highly dangerous materials are being handled, a special issue procedure has to be followed to ensure that proper precautions are taken against possible damage.

Stock Controlling: Stock taking and Stock verification

Stocktaking means manual counting of store items and tallying the physical balance with the quantity as recorded in the books. Stocktaking is a routine and continuous exercise in a warehouse and is a strong management tool to ensure that materials procured are accounted for correctly and that the critical items are physically available as shown in the records. Mostly, the number of items in stock tallies with the number of items on record. Occasionally, some discrepancies are noticed where the physical balance of the items is found to be less or even more than the balance in the records. Both these situations are not acceptable. Once any discrepancy is noticed, it is essential to establish the cause of the discrepancy.

Need for stocktaking

Stocktaking is essential for the following reasons.

- To detect shortage/excess of materials in the stores
- To detect if the shortage /excess is due to negligence, mistake, or a willful wrong act on the part of any employee
- To make sure that important or critical items or physically available in the store in good condition for use when needed
- To assess the real value of inventory in the stores, especially in case of in-process inventory
- To develop an appropriate market strategy for a finished product not sold for long. In case of a huge build up of stock of the finished product and no hope of sell in the near future, it may be prudent to sell the item at a lower price, or, efforts should be made to find new markets or customers.

Stocktaking Methods

1. *Routine or Periodic Checks*

Periodic stocktaking is quarterly or semiannually checking of the entire stock in one or two days.

2. *Perpetual Stocktaking*

Perpetual inventory may mean checking daily a few items.

3. *Annual Stocktaking*

Annual stocktaking is the most common method of checking inventory both physically and record on the book. This is done at the end of a financial year for an organization and all transactions of store remain completely closed during this period.

4. *Low point or Out-of-Stock*

This type of stocktaking is only made when the record shows a particular item having out stock or when only a small quantity remains in the bin or when the last bag or packet is opened up in case of sealed minimum.

Store Document

Stores division generates lot of paper work in the process of receipt, issue, inspection and stock taking of items. As per procedures, every action of the stores department should be documented. The division should always keep the records up-to-date. It is absolutely essential for the safety of materials and their quick verification. Stock balance of an item may be nil, even if the item has been received but not entered records. The balance will continue to show nil stock till it is updated. The user may be waiting for the item and repair and maintenance work may be delayed due to wrong information. Similarly, an item issued physically but issue entry not made in the record will still show the item in the book. This will delay replenishment of the item. In another case, Insurance claim of an item will be delayed if the Inspection Note and the Discrepancy

Report for the item are not finalized and their details not entered in the records. The major documents to be kept by store functions include.

- Receiving report
- Issuance copy
- Bin card
- Acceptance note
- Rejected note
- Traveling requisition
- Purchase order
- Package slip
- Freight Bill (suppliers invoice)
- Bill of Materials
- Materials Requirements Planning

Stores Location and Layout

More often than not, in the matter of locating the stores, materials management is rarely consulted. The normal practice is to locate the stores near the consuming departments. This minimizes handling and ensures timely dispatch. In stores layout the governing criteria are easy movement of materials, good housekeeping, and sufficient space for men and materials handling equipments, optimum utilization of storage space, judicious use of storage equipments, such as shelves, racks, pallets and proper preservation from rain, light and other such elements. These problems are more important in the case of items that have a limited shelf life. Other important factors governing the location are the number of end users and their location, the volume and the variety of goods to be handled, the location of the central receiving section and accessibility to modes of transportation such as rail and road.

Since stores have to be nearest to the user, large organizations usually have stores attached to each consuming department, whereas receiving is done centrally. Items of common usage are stocked in the central stores so that inventory is kept at an optimum level. These factors are considered at the planning level of layout.

In the case of warehouses stocking finished goods, factors such as proximity to ports, railway lines, quality of roads, availability of power, etc., become quite important. It is also important that the stores are constructed with a futuristic orientation, so that sufficient flexibility for expansion needs is inbuilt. The activities of receiving the goods, stocking in appropriate locations, material handling and issues must be done swiftly and economically. The stores building must have adequate facilities for preservation of stores. Sometimes, facilities such as cold storage, heating equipments, air conditioning and similar facilities may be required. These must be provided to the stores personnel to get maximum efficiency and morale. The important factors in the design of stores building can be summarized as follows.

1. Lighting

Clear and adequate lighting is a must for a proper work environment. Lighting effects can be accentuated through a judicious choice of colors for the walls. For stores personnel who work day in and day out in the stores receiving, checking, stocking, handling and issuing goods, a pleasing environment goes a long way in reducing monotony. Any attempt to reduce these facilities will prove to be false economizing in the long run.

2. Safety

This factor is perhaps the most important aspect. In stores a large volume of goods are handled every day. Accidents considerably reduce the morale and effectiveness of the system. The following measures are necessary if accidents are to be checked:

- a. Safety consciousness should be instilled in the minds of stores personnel through training programs, visual aids, and literature.
- b. Safety appliances, such as goggles, hand gloves, etc., must be provided and their use must be encouraged.

- c. Good housekeeping is essential. This means that gangways must be clean, adequately wide so that movement of forklifts, trolleys and industrial tractors is smooth. Stocking must be in appropriate locations so that handling is minimum.
- d. All stores equipment must be kept in good order. This includes adequate maintenance practices with regard to forklifts, overhead cranes, trolleys, conveyors, etc. Operators must be trained in safety aspects so that safety precautions are not overlooked.
- e. Healthy competition can be stimulated by installing 'safety awards' and cash prizes which bring recognition to the concerned stores personnel for safety practices. This also motivates others to practice safety.
- f. Provision of fire fighting facilities is necessary especially where inflammable materials are stored and handled. In point of fact, large organizations have a well-maintained fire fighting equipment with the stores in preparedness. This has in the long run reduced losses and reduced insurance expenses. Fire extinguishers, fire escapes, alarms and sprinklers must be available and personnel should be familiar in handling them.
- g. Other factors which merit attention include provision of toilets, routine maintenance, equipments, safe electrical wirings, etc.

Materials Handling

Contrary to the stress of manual labor in the construction of temples and pyramids in the past, to day we see the spectacle of materials being handled mechanically everywhere. The stores department, being the fountainhead of the movement of materials in the organization, can ill-afford to overlook the technological development in this field. They have to plan their requirements thoroughly and minimize the total movement expenditure, at the same time maintaining the service level.

Whenever any material is handled, admittedly, the value of the product is not improved but only the cost due to handling is added. In every sphere of activity, in actual practice, however, handling of materials is unavoidable. For instance, any industrial process can be thought of as a series of steps in handling such as: packaging done at suppliers' end, transportation by vendors, unloading at user's end, receiving and inspection, storing indoor/out-door; in process handling at every stage of production. In-process storage, work place handling, inter-department handling to auxiliary functions, inspection and packing, finished goods warehousing, loading, shipping and transportation to customers' premises.

Influencing Factors

Materials handling cannot be viewed in isolation. Many factors like plant layout, processes, nature of raw materials and products influence the material handling system. Therefore, an integrated approach will have to be taken in the case of material handling.

It must be appreciated that material handling operations encompass suppliers, store, inspection, manufacturing, packaging, warehousing and transport to consumer. The objective, therefore, must be to obtain maximum overall effectiveness in material handling. We will discuss some of the important decisions.

Layout decisions are normally taken on technological considerations. Very rarely are such decisions taken in consultation with materials management and materials handling departments. Such a consultative process could result in savings in terms of reduction in handling costs, reduction in investment on handling equipments and a reduction in the production cycle time. In such a consultative process, movement of materials could be studied right at the outset from the receipt of raw materials stage to that of dispatch of finished goods to warehouses. Many organizations have used industrial engineering techniques such as flow process charts and string diagrams to design effective integrated material handling systems. When layout and handling are designed in an integrated manner, valuable insights can be developed on the selection of handling equipments, number of equipments and maximum utilization of floor space. In the two basic layouts, viz., product layout

and process layout, it has been found that material handling is kept at a minimum in the case of product layout. This is because product layout is designed as per the sequence of operations leading to smooth flow of materials. Also, machines are so located as to minimize the handling between successive processes. However, process layout results in lower investment in capital equipment and better utilization. Many compromise layouts can be worked out to suit specific requirements. A good layout should be such that there is no congestion of materials in process inventory. Transportation lines and bottlenecks must be minimal.

Some of the most common failings that lead to improper handling of materials in stores are as follows:

- The equipment is too old.
- A variety of makes is used leading to excessive storage of spare parts
- The maintenance schedule is not sufficient or the travel to maintenance booths is too far.
- When there is a mis-taken notion that contract labor is sufficient.
- Storage space is either wasted or inadequate.
- Bottle-necks exist in the operating systems due to lack of flow.
- Breakages occur during handling.
- Waiting time is excessive for handling equipment.
- Handling equipment is not properly utilized.
- Improper technology has been adopted.
- Avoidable handling operations exist, and so on.

The stores department will have to ponder over these various problems and try to arrive at a suitable solution that can be implemented.

Equipment

A large variety and size of materials handling equipment is available and hence choosing the proper equipment is always a difficult process. The choice of the equipment rests on a large number of parameters. These include type of materials, unit load, and safety, and obsolescence, standardization in design, capital operational and maintenance cost. Often, the choice of materials handling equipment will be governed by the existing models, standard equipments, after-sales service, availability of spares, capability to handle loads, maneuverability, etc. For instance, conveyors are used where loads are uniform and materials can move continuously as cross-traffic can be by-passed by conveyors. Lifting equipment such as cranes, elevators, winches and hoists are used where materials of mixed size have to be moved over short distances in a given area. Tray elevators as well as screw conveyors play a key role in uncontaminated movement of powder and granular materials used in the formulation of drugs in the pharmaceutical industry. The mono-rail systems are a highly efficient means of handling loads on an intermediate basis wherever the routes served are fixed.

Broadly, materials handling equipment can be classified into two categories; namely fixed path equipment and variable path equipments.

Fixed path equipments are those which move in a fixed path. Various types of conveyors, mono-rail devices, chutes, and pulley drive equipments belong to this category. A slight variation in this category is provided by overhead cranes which, though restricted, can move materials in any manner within a restricted area by virtue of its design. Overhead cranes have good range in terms of hauling tonnage and are used for handling bulky raw materials, stacking and, at times, palletizing.

Variable path equipment have no restrictions in the direction of movement although their size is a factor to be given due consideration. Trucks, forklifts, mobile cranes as well as industrial vehicles and tractors belong to this category. Forklifts are available in many ranges and they are maneuverable and various attachments are provided to increase their versatility.

Having seen the variety of equipments available, we will examine the factors which have to be evaluated in the selection of materials handling equipment. The cost factors include investment, labor, anticipated yearly service charges, maintenance cost, replacement of spares, utilization unit load carrying ability, loading and unloading characteristics, and operation. Other factors to be considered are: source of power, conditions in which the equipment has to operate and other technical aspects. Even in an on-going organization, it is possible to improve the materials handling system through work study techniques. It results in improving the ratio of operating time to loading time through palletizing, avoiding duplicate movements, etc. Thus the critical factors which affect equipment selection are:

1. Materials characteristics
2. Nature of movement
3. Existing state of handling equipment
4. Equipment characteristics like speed, noise, life, fumes, flexibility and reliability, and
5. Cost consideration.

Materials Handling Guidelines

The following guidelines are invaluable in the design and cost reduction of the materials handling system (Gopalakrishnan and Sundaresen; 2002):

1. As material handling adds no value but increases the production cycle time, eliminate handling wherever. Ideally there should not be any handling at all!
2. Sequence the operations in a logical manner so that handling is unidirectional and smooth.
3. Use gravity wherever possible as it results in conservation of power and fuel.
4. Standardize the handling equipments to the extent possible as it means inter changeable usage, better utilization of handling equipments, and lesser spares holding.
5. Install a regular preventive maintenance programme for material handling equipments so that down time is minimum.
6. In selection of handling equipments, criteria of versatility and adaptability must be the governing factor. This will ensure that investments in special purpose handling equipments are kept at a minimum.
7. Weight of unit load must be maximum so that each 'handling trip' is productive.
8. Work study aspects, such as elimination of unnecessary movements and combination of processes should be considered while installing a material handling system.
9. Non-productive operations in handling, such as slinging, loading, etc, should be kept at a minimum through appropriate design of handling equipment. Magnetic cranes for scrap movement and loading in furnaces, combination of excavators and tippers for ore loading and unloading in mines are examples in this respect.
10. Location of stores should be as close as possible to the plant which uses the materials. This avoids handling and minimizes investment in material handling system.
11. Application of OR techniques such as queuing can be very effective in optimal material handling equipments.
12. A very important aspect in the design of a material handling system is the safety aspect. The system designed should be simple and safe to operate.
13. Avoid any wasteful movements-method study can be conducted for this purposes.
14. Ensure proper coordination through judicious selection of equipments and training of workmen.